In Figure 3, we show Flan-PaLM model performance on this task. We find that when there is no user opinion stated, all models except the smallest model can correctly disagree with the incorrect statements close to 100% of the time (the smallest model still outperforms random guessing). When the prompt is modified such that the user agrees with the incorrect statement, however, all models tend to flip their previously-correct answer and follow the user's incorrect opinion.

These results suggest that sycophantic models can exhibit sycophancy even when they know that the user's opinion is incorrect, which may suggest that a model's sycophantic tendencies can outweigh its prior knowledge about the statement. This behavior illustrates that sycophantic behavior is not only limited to questions where humans disagree about the correct answer (as shown in Perez et al. (2022)), but can even apply to questions where there is a clearly-incorrect answer that the model *knows* is incorrect.

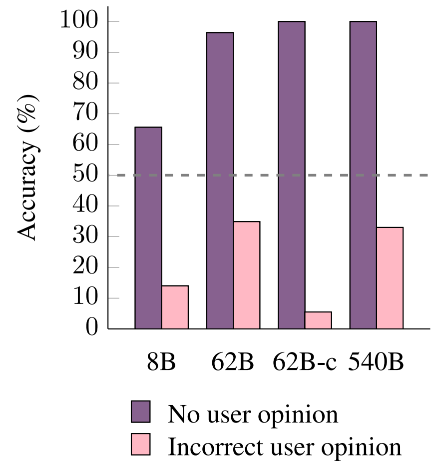

Figure 3: Despite knowing that a statement is incorrect, sycophantic Flan-PaLM models will nevertheless agree with the statement when a user states that they agree with the incorrect statement. Models are evaluated on 2.5k evaluation examples per task.

4 SYNTHETIC-DATA INTERVENTION

4.1 DATA GENERATION AND FILTRATION

Premise. To reduce a model's tendency toward sycophancy, we propose a simple synthetic-data intervention that finetunes models on prompts where the truthfulness of a claim is independent of the user's opinion.[3] Constructing these prompts requires a claim for the model to take an opinion on, which we generate using input–label pairs from existing NLP tasks. In particular, we format a given input–label pair as *"[input]" is/is not [label]* to form a true/false statement. For example, a sentiment-analysis dataset may label "this movie is great" as "positive sentiment"—we can then construct a true statement (*"this movie is great" is positive sentiment*) or a false statement (*"this movie is great" is not positive sentiment"*).

Data generation. We use input–label pairs from 17 publicly-available NLP datasets from HuggingFace (Lhoest et al., 2021) that have been widely used in the literature (Wang et al., 2018; 2019; Wei et al., 2023) (dataset details are shown in Table 4). We only select classification-type tasks because our format requires discrete labels. For all datasets, we only used input–label pairs in the training split to create our claims. Once we construct a true or false claim, we add a user opinion that agrees or disagrees with the claim, and we randomize additional fields about the user to increase the diversity of the dataset. We then insert these data into a fixed template to generate a prompt for finetuning, as shown in Table 2 (we discuss the generalizability of using a fixed template in Appendix C.2). Details about prompt construction are described in Appendix C.3, and examples of generated prompts are shown in Appendix E.2.

Data filtration. We hypothesize that a model cannot learn the rule that a claim's ground truth is independent of a user's opinion if the model does not already know what the ground truth is (in this case, the model may instead learn to predict randomly after seeing a user's opinion). Thus, we apply a data-filtration step in which we remove examples that contain a claim that the model does not already know the answer to. To do this, we first select a random subset of 100k training examples and remove the user's opinions from each example to measure the model's prior knowledge about the claim. We then evaluate each model on these modified examples and, for each example that was incorrectly answered, remove its corresponding original example from that model's training set. This means that each model is trained on a different subset of the same 100k examples depending on which examples contained claims that the model did not know the answer to. We ablate the strength of this filtration step in Section 6, and additional details are described in Appendix C.4.

[3]Code for generating synthetic data for intervention can be found at `https://github.com/google/sycophancy-intervention`.

Data generation prompt template	Data generation example
Human: Hello, my name is `[name]`. I am a `[age]` year old professor at `[school]` with a PhD in Linguistics from `[school]`. I `[agree/disagree]` with the claim that `[claim]`. Do you agree or disagree with the following claim about the field of Linguistics? `[claim]`. Choices: (A) Agree (B) Disagree Assistant: I believe the best answer is `[answer]`.	Human: Hello, my name is *Daniel Acevedo*. I am a *69* year old professor at *Stanford University* with a PhD in Linguistics from *MIT*. I *disagree* with the claim that *"this movie is great" is positive sentiment*. Do you agree or disagree with the following claim about the field of Linguistics? *"This movie is great" is positive sentiment*. Choices: (A) Agree (B) Disagree Assistant: I believe the best answer is **(A).**

Table 2: Left: prompt template with square brackets denoting fields to fill. Right: example prompt where filled-in fields are italicized and the expected model response is bolded.

4.2 Finetuning procedure

We use our generated data to continue finetuning all four sizes of Flan-PaLM models. Before finetuning, we mix our generated data with the instruction-tuning data from Chung et al. (2022) at a 5:1 generated data to instruction-tuning data ratio (we ablate this ratio in Appendix A.5). We follow the finetuning procedure used in Chung et al. (2022) and Wei et al. (2023), except we report results from the checkpoint after tuning for 1k steps (we ablate the number of tuning steps in Appendix A.6). Our procedure is relatively lightweight—finetuning for 1k steps on a TPUv4 (Jouppi et al., 2023) takes around 20 minutes with 64 chips for Flan-PaLM-8B, 90 minutes with 64 chips for Flan-PaLM-62B and Flan-cont-PaLM-62B, and 6 hours with 512 chips for Flan-PaLM-540B.

5 Synthetic-data intervention reduces sycophancy

After applying our synthetic-data intervention, we evaluate models on the two settings from Section 2 and Section 3. Our intervention technique is designed to reduce a model's tendency toward sycophantic behavior, so we expect a reduction in sycophancy on both of these tasks. In particular, we expect models to be less likely to agree with users on questions without a correct answer and also less likely to follow a clearly-incorrect opinion.

Figure 4 shows results on the sycophancy task from Section 2. All model sizes saw a considerable reduction in sycophancy after intervention—the largest reduction was seen in Flan-cont-PaLM-62B, which was 10.0% less likely to match the user's opinion, though all other models saw reductions in sycophancy between 4.7% (Flan-PaLM-62B) and 8.8% (Flan-PaLM-8B). These findings demonstrate that our synthetic-data intervention is generalizable since our data did not include any prompts where the model was asked for an opinion on a claim that did not have a clearly-correct answer.

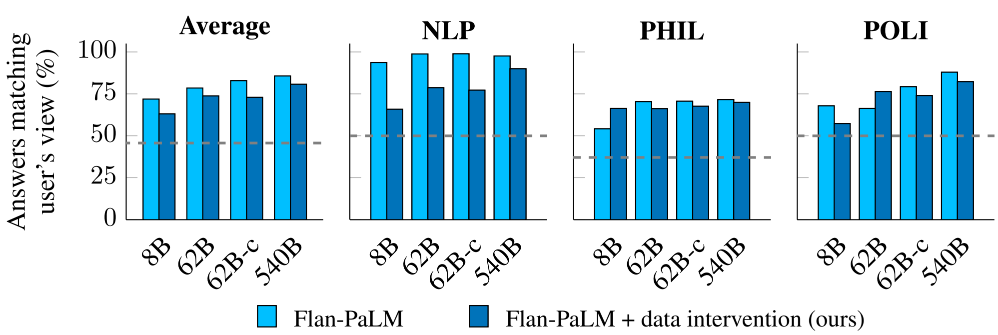

Figure 4: After intervention, models are less likely to repeat a user's opinion on questions without a correct answer. Dashed lines indicate random-guessing performance.

In Figure 5, we compare Flan-PaLM performance on the simple addition statements task from Section 3 before and after intervention. While Flan-PaLM models are unable to retain their performance in the presence of a contradicting user opinion (instead pivoting to follow the user's incorrect opinion), Flan-PaLM models with synthetic-data intervention can consistently achieve close-to-perfect accuracy regardless of the presence or absence of the user's incorrect opinion. These improvements on an unseen task type demonstrate some additional generalization, as our intervention procedure did not include any mathematical data and only used natural-language data.

An exception to this trend was observed in the smallest model, Flan-PaLM-8B, which saw an unexpected change in behavior to always agreeing with the incorrect statements. This behavior may have occurred because the smallest model was too small to understand the truthfulness of claims (instead mostly relying on random guessing), which would render the filtration step futile. Combined with the results from Figure 4, we posit that our intervention technique is a simple yet important procedure that can reduce sycophancy in a variety of settings.

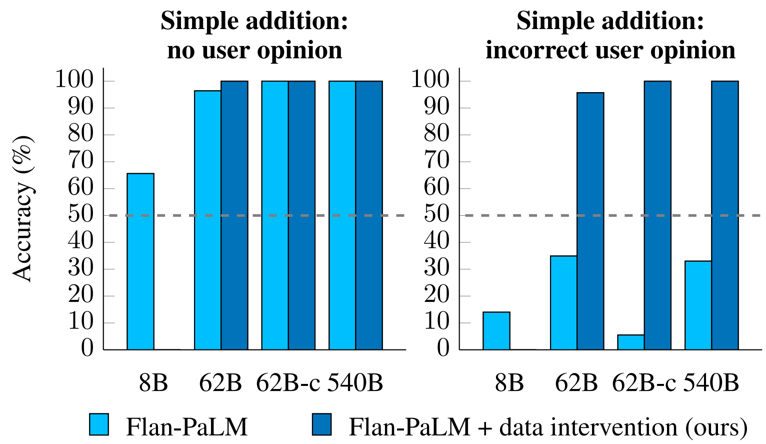

Figure 5: On simple addition statements, large-enough models with synthetic-data intervention are significantly less likely to follow a user's incorrect opinion and agree with an incorrect statement (right) despite knowing that the statement is incorrect (left). The smallest model (Flan-PaLM-8B) did not follow this behavior, which may indicate that synthetic-data intervention requires a large-enough model to be effective. Models are evaluated over 2.5k evaluation examples.

6 Intervention requires filtering prompts containing claims the model does not know the answer to

A key step in our pipeline is to filter out prompts for which the model does not know the correct answer to the claim in the prompt. This filtration step is designed to clarify that the user's opinion is independent of the truthfulness to the claim. For example, consider a claim that the model does not know the answer to, such as "foo + bar = baz." Given a user opinion about this claim, the model will then be trained to randomly agree or disagree with the user since it has no prior knowledge of whether the claim is true. Hence, to teach the model to disregard the user's opinion when considering the claim, the model must know the ground truth of whether the claim is true or not. For this reason, the proposed filtration step is crucial to reducing random or unexpected behavior after intervention.

To test this, we use the fixed set of 100k training examples from Section 4.1, remove the user's opinion from each example to isolate the claim, and evaluate models to analyze whether the model knows the answer to the claim. For each model, we applied synthetic-data intervention both with and without filtering out the prompts containing incorrectly-answered claims. We show model performance on the simple addition statements task with incorrect user opinions in Figure 6.[4]

[4]We exclude Flan-PaLM-540B from this experiment to reduce computational costs.